PROBLEM: RESERVE (contributed by F. R. Salvador)

Demonstrate that you can apply what we learned on arrays and strings by defining 5 functions that will be used by a Venue Reservation System with the following descriptions:

- There is only one venue, and the reservation is for this year 2025 only (for simplicity).
- The venue can be reserved in the morning and afternoon (actual reservation time is not specified to simplify the problem).
- There can only be at most 1 reservation in the morning, and 1 reservation in the afternoon on each day of the month.
- The reservation information are recorded in a 2D array named **RESERVE** which is made up of 12 rows and 31 columns.
- The rows represent the 12 months of the Year, and the columns represent the days of the month (at most 31 days).
- A **RESERVE** array element is accessed as **RESERVE[month_index][day_index]** where **month_index** is from 0 to 11, and **day_index** is from 0 to 30.
    - month_index of 0 corresponds to the month of January, month_index of 1 for February, …, month_index of 10 for November and month_index of 11 for December.
    - day_index of 0 is for the 1st day of the month, day_index of 1 is for the 2nd day of the month, and so on…
      Example #1: **RESERVE[0][0]** contains the reservation info for January 1st
      Example #2: **RESERVE[11][24]** contains the reservation info for December 25
- Each array element in the **RESERVE** array contains a string made up of at most **50** characters *excluding* the null byte.
- Each string complies with the following format shown in color for easier understanding:

  "NAME-NUMBER#NAME-NUMBER"

    - The red-colored portion contains the name, followed by a dash(or minus) character, followed by the contact number of the client for the morning reservation.
    - The blue-colored portion contains the name, followed by a dash (or minus) character, followed by the contact number of the client for the afternoon reservation.
    - A crosshatch character **#** serves as a separator to distinguish the morning from the afternoon reservation information.
    - The contact number is made up of 5 digits.
    - There are NO white spaces, i.e., blanks or tabs in the string.

Study/understand the following example strings:

Example #1: "**ENZO-91712#MIKEE-91867**" this means the venue is reserved in the morning by ENZO with contact number 91712, and in the afternoon by MIKEE with contact number 91867.

Example #2: "**ASHLEY-90789#ASHLEY-90789**" this means the venue is reserved both in the morning and afternoon by ASHLEY with contact number 90789.

Example #3: "**ANDREW-99845#-**" this means the venue is reserved in the morning by ANDREW with contact number 99845, and that there is no reservation in the afternoon

Example #4: "**-#APPLE-99581**" this means that there is no reservation in the morning, and that there is a reservation in the afternoon by APPLE with contact number 99581

Example #5: "**-#-**" this means there are no reservations both in the morning and in the afternoon

The figure below shows a visualization of an example **RESERVE** array. It shows that **RESERVE[0][0]** contains the string "**ENZO-91712#MIKEE-91867**", **RESERVE[1][2]** contains the string "**ANDREW-99845#-**", **RESERVE[1][27]** contains the string"**-#APPLE-99581**", **RESERVE[4][2]** and most of the other elements contain the string "**-#-**". The colons represent other cells that cannot be shown anymore due to space constraints.

		day_index							
		0	*1*	*2*	*:*	*:*	*27*	*:*	*30*
month_index	*0*	ENZO-91712#MIKEE-91867	ASHLEY-90789#ASHLEY-90789	-#-	:	:	-#-	:	-#-
	1	-#-	-#-	ANDREW-99845#-	:	:	-#APPLE-99581	:	-#-
	2	-#-	JENNIFER-90301#-	-#-	:	:	-#-	:	-#-
	3	-#-	-#-	-#-	:	:	-#-	:	-#-
	4	-#-	-#-	-#-	:	:	-#-	:	-#-
	5	-#-	-#-	-#-	:	:	-#-	:	-#-
	6	-#-	-#-	-#-	:	:	-#-	:	-#-
	7	-#-	-#-	-#-	:	:	-#-	:	-#-
	8	-#-	-#-	-#-	:	:	-#-	:	-#-
	9	-#-	-#-	-#-	:	:	-#-	:	-#-
	10	-#ANDY-91101	-#-	-#-	:	:	-#-	:	-#-
	11	-#-	-#-	-#-	:	:	-#-	:	-#-

SOME IMPORTANT HINTS that may be of help to you:

- If the first character of the string **`RESERVE[i][j]`** is a **`'-'`** (dash or minus character), then it means that there is no reservation (yet) in the morning. Otherwise, there is already an existing morning reservation.
- Similarly, if the last character (before the null byte) of the string **`RESERVE[i][j]`** is a **`'-'`**, then it means that there is no reservation (yet) in the afternoon. Otherwise, there is already an existing afternoon reservation.
- Let **`str`** be a string variable. The first character is accessed as **`str[0]`**. The last character (before the null byte) is accessed as **`str[strlen(str) - 1]`**.
- Remember that the crosshatch character **#** serves as a separator to distinguish the morning from the afternoon reservation information. Knowing the array index where **#** is stored in a string *can* be of use in formulating some of the algorithms needed to solve this exam.

Your TASK is to edit the accompanying skeleton file **`RESERVE-LASTNAME.c`** file and define the body of the 5 functions listed below. The specific requirements for each function are described in the comments that appear before the function header.

1. **`int SearchChar(char key, char *string)`**
2. **`int CheckAvailability(Str50 RESERVE[][NDAYS], int month_index, int day_index)`**
3. **`void Who(Str50 RESERVE[][NDAYS], int month_index, int day_index, int flag, Str50 answer)`**
4. **`int Book(Str50 RESERVE[][NDAYS], int month_index, int day_index, int flag, char *name, char *number)`**
5. **`void CountReservations(Str50 RESERVE[][NDAYS], int COUNT[])`**

Note that **`Str50`** is an alias declared as: **`typedef char Str50[51];`** and **`NDAYS`** is a macro definition for 31. Please refer to the accompanying **`reserve.h`** file.

FILES PROVIDED: You are given the following files for this problem

- **`RESERVE-LASTNAME.c`** -- skeleton file which contains some initial code that you'll need to complete
- **`main.c`** -- contains the **`main()`** function for testing purposes; make sure to read & understand this file
- **`reserve.h`** -- header file; make sure to read & understand this file
- **`INPUT.txt`** -- example input data file used to initially populate the **RESERVE** array.
- **`EXPECTED.txt`** -- contains the expected results when the **`exe`** file is executed using the data stored in **`INPUT.txt`** file

DELIVERABLES: Submit/upload two files before the Canvas submission deadline

1. **`RESERVE-LASTNAME.c`** -- this file contains your solution (i.e., C source code)
2. **`ACTUAL-LASTNAME.txt`** -- this is the actual output redirected text file produced by your own exe file. You can produce this file by running your a.exe file with OUTPUT redirection in the command line interface as follows:

   prompt> **`a > ACTUAL-LASTNAME.txt`** **(Press Enter)**

Make sure to rename these files using your own last name. For example, if your last name is SANTOS, you must upload your files as RESERVE-SANTOS.c and ACTUAL-SANTOS.c.

TESTING & SCORING:

- Your program will be tested using a different input data set. Thus, your solution should not be fixed to the given sample data only.
- Each correct function definition will be given **10 points** each. Thus, the maximum total score will be 50/50.
- **A function that has a syntax/compilation error will result in a score of 0.**
- **A function that does not produce any of the required output (for example, due to runtime error) will result in a score of 0.**
- **A function should be defined using ONLY the ideas/concepts/syntax/library functions covered so far in CCPROG1 and in Chapters 1 and 2 of our CCPROG2 Course Notes. Non-compliance will result in a score of 0 in the worst case scenario for the offending function definition.**

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